4 The probability distribution of the discrete random variable X is given in Fig. 4.

r	0	1	2	3	4
$P(X = r)$	0.2	0.15	0.3	k	0.25

Fig. 4

(i) Find the value of k. **[2]**

X_1 and X_2 are two independent values of X.

(ii) Find $P(X_1 + X_2 = 6)$. **[3]**